

Appendix G — UIP v4: Inaccessible-Memory Chain Experiment

UIP Phase 1 Review Packet · Independent Experiment · Date run: 2026-07-15 · Status:

PREREGISTERED, EXECUTED, FALSIFIED

1. Preregistration

System. Three qubits in a chain $A \rightarrow M \rightarrow B$. Partial-SWAP coupling $A \leftrightarrow M$ ($R_1 = 0.30$) and $M \leftrightarrow B$ ($R_2 = 0.30$). Independent dephasing on all three qubits ($\gamma_A = 0.06$, $\gamma_M = 0.06$, $\gamma_B = 0.015$). Recirculation correction toward $|+\rangle$ (as in Appendix A) is allowed **only on A and M** — the memory qubit B is untouchable by construction. Correction budget: one qubit per cycle, gain $g = 0.5$. Duration: $T = 200$ cycles.

Policy under test (uip_v4). Each cycle, correct the qubit with the larger chain-discounted deliverable inheritance toward B:

$$I_2(A) = R_1 \cdot R_2 \cdot C(A) \cdot \tau^2 \quad I_2(M) = R_2 \cdot C(M) \cdot \tau$$

Baselines. always-A, always-M, alternating, random.

Preregistered predictions.

- **P1:** With a noise attack alternating between A and M (heavy dephasing switching every 25 cycles), uip_v4 beats all four baselines in mean $C(B)$.
- **P2:** Under static noise, uip_v4 performs within 3% of the best static baseline.
- **Falsification condition:** uip_v4 underperforms any baseline by more than 5%.

2. Results

Policy	Mean $C(B)$ — attack regime	Mean $C(B)$ — static noise
always-A	0.2664	0.4571
always-M	0.3742	0.4606
alternate	0.4431	0.6145
random	0.4145	0.5854
uip_v4	0.3742	0.4606

- **P1: FAIL** — uip_v4 underperformed the alternating baseline by 15.54% (threshold: 5%).
- **P2: FAIL** — uip_v4 was 25.04% below the best static baseline.
- **Preregistered falsification condition met: FALSIFIED.**

3. Diagnosis

The chain-discounted inheritance rule collapsed into “always correct M” — its results are numerically identical to the always-M baseline in both regimes. The $R_1 \cdot R_2$ discount meant qubit A almost never won the comparison, so the nominally adaptive policy was never adaptive in practice.

The alternating baseline dominated because chain transport is a *pipeline-maintenance* problem: both links require periodic refreshing, and any greedy single-target rule starves one link of the chain.

4. Conclusion

This is the third consecutive falsification (Appendices B, F, G) of inheritance-guided control, now including its most favorable possible setting — a target that cannot be corrected directly. The scientific statement Phase 1 supports is therefore sharp:

The UIP inheritance quantity, in every form tested (I , I_2 , chain-discounted I_2), does not predict optimal correction allocation. Simple resource-spreading policies dominate.

Appendix A (single-qubit recirculation with steady-state prediction within 2%) remains the framework’s only pass.

Reported in full per the project’s open-failure policy.

5. Methodological Note

Predictions P1 and P2 were specified before execution but were not publicly timestamped prior to the run. The accompanying GitHub commit (repository: `uip-phase1-testbeds`) establishes the public record. Future experiments — including all Phase 2 work — adopt a commit-preregistration-first workflow: preregistration is committed and timestamped publicly before any simulation is executed.

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